Balloon-Borne Profiling of Global UV-A Attenuation over the Kota Region, Rajasthan, India: Effective Optical Depth, Satellite Aerosol Context, and Instrument Performance from a Low-Cost High-Altitude Flight

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Abstract

We report a single-flight case study of downwelling global (hemispherical) UV-A irradiance measured from a low-cost high-altitude balloon (HAB) launched from Sultanpur, Kota district, Rajasthan, India (25.298°N, 76.192°E) on 5 October 2025. The payload carried a factory-calibrated Apogee SU-202 UV-A radiometer (305–390 nm) as the primary instrument, alongside a pressure–temperature–humidity sensor, Geiger–Müller (GM) counter, ozone sensor, GPS, and thermal imager for context. The balloon reached ≈17.6 km (pressure altitude) at a mean ascent rate of 3.0 $\mathrm{m\,s^{-1}}$. UV-A irradiance increased from 7.6 $\mathrm{W\,m^{-2}}$ at launch (solar zenith angle 78°) to 79.2 $\mathrm{W\,m^{-2}}$ at apogee, with a propagated per-sample uncertainty of ±5.8% (median, k=1) dominated by the ±5% factory calibration term; gyro logs from the onboard camera show the payload swung by only 0.9° (median tilt) during ascent, so the tilt term in the budget is measured rather than assumed. Because calibration systematics cancel in irradiance ratios, a Langley-type retrieval anchored at apogee yields an *effective* vertical optical depth for global UV-A of $\tau_{\mathrm{eff}} = 0.233 \pm 0.023$, with attenuation concentrated below ≈2.5 km, consistent with the ERA5 boundary-layer height (1.4 km). By contrast, the direct-beam extinction expectation for the same column—Rayleigh scattering ($\tau_R = 0.551$ at 360 nm) plus MAIAC satellite aerosol optical depth extrapolated to 360 nm via the Ångström exponent ($\mathrm{AOD}_{360} = 1.19 \pm 0.21$)—is $\tau_{\mathrm{ext}} = 1.74 \pm 0.21$. The measured ratio $\eta = \tau_{\mathrm{eff}}/\tau_{\mathrm{ext}} = 0.134 \pm 0.021$ quantifies the resilience of hemispherical UV-A irradiance to a moderately turbid, non-absorbing atmosphere (TROPOMI UV aerosol index −0.19): scattered photons are largely returned to the downwelling stream rather than removed. A pre-launch surface reading agrees with the independent NASA POWER satellite-derived surface UV-A product to within ≈25%, inside the combined uncertainties of the two estimates. Surface air quality during the flight was moderate (CPCB $\mathrm{PM}_{2.5}$ 24–29 $\mu\mathrm{g\,m^{-3}}$ daily mean at two stations). We include a full error-propagation budget, document three instructive instrument failures (GM-tube corona discharge above 4.7 km, a non-responsive ozone channel, and GPS time-stamp freezing), and provide recommendations for repeat flights. The dataset demonstrates that research-grade radiometry with quantified uncertainties is achievable on student-class HAB platforms in regions where reference monitoring is sparse.

Keywords: high-altitude balloon; UV-A irradiance; aerosol optical depth; MAIAC; error propagation; India; low-cost sensors; atmospheric attenuation

1 Introduction

Satellite retrievals of aerosol optical depth (AOD) are the backbone of air-quality assessment in regions with sparse ground monitoring, but they require local validation and context to be interpreted correctly [1, 2, 3]. In India, long-term satellite $PM_{2.5}$ products [4, 5] and sensor-evaluation studies [6] have grown rapidly, yet vertically resolved in-situ observations remain rare outside major campaigns. Low-cost balloon platforms have been shown to close part of this gap: tethered and free balloons carrying miniature particulate and radiation sensors have produced useful vertical profiles in Paris [7], on stratospheric student flights [8], and in agricultural-burning regions [9], while sensor-package designs for boundary-layer profiling are maturing [10].

This paper asks a deliberately narrow question with one flight's data: *how strongly does a real, moderately turbid subtropical atmosphere attenuate downwelling global (direct + diffuse) UV-A irradiance, and how does that compare with the direct-beam extinction implied by satellite AOD?* The distinction matters. Extinction optical depth, the quantity retrieved by sun photometers and satellites, describes the attenuation of the *direct beam* only. A hemispherical (180° field-of-view) radiometer measures global irradiance, which is partially replenished by multiple scattering; for a high-single-scattering-albedo aerosol, most scattered UV photons remain in the downwelling stream. Quantifying the gap between the two—the ratio we call η—with propagated uncertainties is the central contribution here, together with a documented uncertainty budget and honest reporting of instrument failures, which we believe is of practical value to the growing community of student and citizen-science HAB programmes [11, 12].

The Kota region is a suitable target for such a study. CPCB air-quality records [21] show that a large majority of post-monsoon days in the region (≈80%) rate "moderate" or poorer on the national AQI scale, driven mainly by industrial, construction, and vehicular sources, yet the region has no routine vertically resolved atmospheric observations. The flight analysed here took place in the first week of the post-monsoon season, on a day whose surface air quality (Section 4.5) was typical of that regime.

2 Flight, instruments, and data

2.1 Mission summary

The Akāsha Jyoti mission was originally planned from the ICAR-VPKAS and ARIES campuses near Nainital, Uttarakhand, under DGCA approval with an AAI-issued NOTAM; persistent rain during the approved window forced a relocation, and the flight was executed from Sultanpur, Kota district, Rajasthan (25.298°N, 76.192°E, ∼270 m a.s.l.) on 5 October 2025. The payload powered on at 07:14 IST and lifted off at ≈07:21 IST. The balloon reached apogee at 08:51 IST at 81.4 hPa — 17.6 km ISA pressure altitude (≈16.9 km geometric) — at a mean ascent rate of 3.0 $\mathrm{m\,s^{-1}}$, and the record ends after parachute descent at 09:57 IST. The payload logged at ≈7 s cadence, producing 1,444 clean records. Over the ascent the solar zenith angle (SZA) decreased from 78.3° to ≈47°. The balloon drifted ≈70 km east-northeast during the flight (Fig. 1); the Kota city CPCB stations used for surface context lie ≈40 km southwest of the launch point, inside the 50-km satellite buffer. Onboard footage shows a partly cloudy morning: scattered boundary-layer cumulus and a thin, milky mid-level layer visible from above, with deep blue sky overhead from the mid-troposphere upward. The smoothness of the measured irradiance profile suggests the balloon–sun path was mostly cloud-free, but the retrieved effective optical depth is an *all-sky* quantity and may include residual broken-cloud contributions (Section 6).

2.2 Payload

Table 1 summarises the payload. The primary instrument is an Apogee SU-202 amplified UV-A radiometer: spectral range 305–390 nm, output 0–2.5 V over 0–100 $\mathrm{W\,m^{-2}}$ (calibration factor 0.04 $\mathrm{W\,m^{-2}\,mV^{-1}}$), calibration uncertainty ±5%, repeatability <0.5%, non-linearity <1%, temperature response <0.1% $\mathrm{°C^{-1}}$, directional (cosine) error ±2% at 45° and ±5% at 75°, operating range −40 to +70°C [19]. The remaining channels provide context and are treated as secondary.

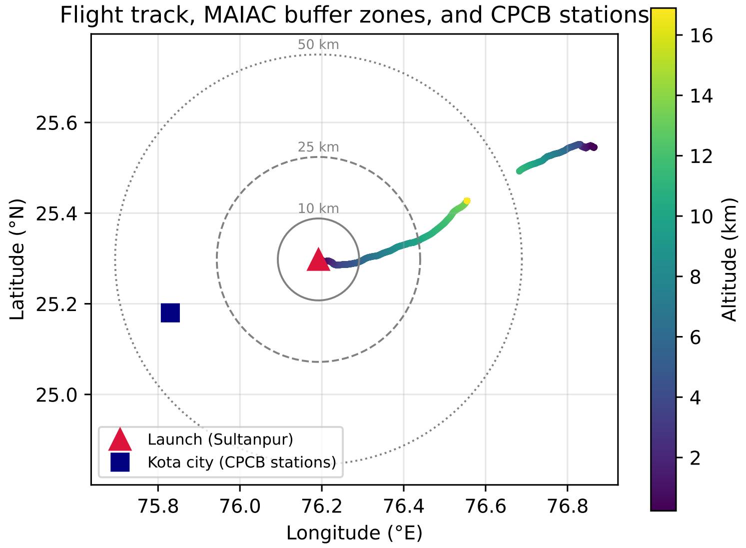

Figure 1: Flight track coloured by altitude, with the 10/25/50-km MAIAC AOD buffer zones around the Sultanpur launch site (triangle) and the location of Kota city with its two CPCB monitoring stations (square).

Table 1: Payload channels, roles, and status during the flight.

Channel	Instrument class	Role	Status
UV-A irradiance	Apogee SU-202 radiometer	**Primary**	nominal throughout
P, T, RH	BME280-class combined sensor	altitude, met	nominal (T payload-internal)
Ionising counts	GM counter (CPM)	cosmic-ray context	valid <4.7 km; HV failure
O_3	semiconductor gas sensor	context	non-responsive (0.00)
GPS	u-blox-class receiver	position, time	position ok; time froze
Video camera + IMU	Runcam camera, 1-kHz gyro/accel	attitude, documentation	nominal throughout
Thermal imager	IR pixel array	qualitative	not analysed here

2.3 Timekeeping correction

GPS time-stamps froze intermittently at altitude (a known behaviour of consumer GPS modules on HAB flights), while the logger's real-time clock (RTC) ticked steadily. All times used here are RTC times anchored to the first valid GPS fix (IST). Solar position was computed per sample with the NOAA solar geometry algorithm using GPS coordinates. The timeline is independently corroborated by the onboard camera's 1-kHz IMU log, in which the release jerk, the balloon-burst onset of violent rotation, and the 18.6-g landing impact appear with relative spacings that match the RTC timeline to within ≈1 minute.

2.4 Payload attitude from the camera IMU

The action camera recorded continuous 1-kHz gyroscope and accelerometer logs (GyroFlow format) for the whole flight, providing a free, quantitative record of payload attitude. During ascent the payload rotated slowly about the suspension axis (median angular rate 0.72 rad s^{-1}, ≈6.8 rpm) and swung very little: the tilt of the low-passed gravity vector from its slow mean had a median of 0.9°, a 95th percentile of 2.6°, and a 99th percentile of 3.9°. Because the payload completes roughly 18 rotations within each 500-m altitude bin, the first-order (azimuth-dependent) cosine error of a tilted sensor averages toward zero, and the residual swing noise appears in the within-bin scatter that the uncertainty budget already propagates (Section 3). The tilt allowance used below is therefore grounded in measurement rather than assumption. Descent under parachute was rougher (median tilt 1.4°, 95th percentile 5.6°), one more reason descent data are excluded from the retrieval.

2.5 Ancillary public data

Four public datasets from the flight day provide context: (i) MAIAC MCD19A2 AOD at 470 and 550 nm [16], extracted along the flight track (0.781 and 0.610 respectively) and in buffers of 10–100 km around the launch site; (ii) Sentinel-5P TROPOMI [17] column products along the track (O_3 281 DU; NO_2 23 μmol m^{-2}; SO_2 0.21 mmol m^{-2}; CO 0.029 mol m^{-2}; UV aerosol index −0.19); (iii) ERA5 reanalysis [18] (boundary-layer height 1,411 m, cloud-base height 1,290 m, 2-m temperature 306.4 K); (iv) CPCB continuous ambient air-quality data from the two Kota stations (Nayapura; Shrinath Puram), reporting 4-hourly $PM_{2.5}$, PM_{10}, NO_2, NH_3, CO and SO_2 [21]; and (v) the NASA POWER SYN1DEG-based hourly surface UV-A product at the launch coordinates [20], used as an independent cross-check of the radiometer.

3 Uncertainty budget and error propagation

3.1 Per-sample irradiance

Irradiance is $E = cV$ with $c = 40$ W m^{-2} V^{-1}. Uncertainty terms combine in quadrature:

$$\frac{\sigma_E}{E} = \sqrt{u_{\text{cal}}^2 + u_{\text{lin}}^2 + u_{\text{rep}}^2 + u_{\text{cos}}^2 + \left(u_T\,|T-25|\right)^2 + \left(\frac{\sigma_q}{V}\right)^2}, \tag{1}$$

with $u_{\text{cal}} = 5\%$, $u_{\text{lin}} = 1\%$, $u_{\text{rep}} = 0.5\%$, $u_{\text{cos}} = 2\%$ (5% for SZA $> 72^\circ$), $u_T = 0.1\%\,^\circ\text{C}^{-1}$, and quantisation $\sigma_q = 0.01/\sqrt{12}$ V from the logged resolution. Over the flight the median total uncertainty is ±5.8% (k=1; worst case 9.1% at the low-signal launch condition), of which only 2.1% is random (Table 2).

Table 2: Error budget for UV-A irradiance and derived quantities.

Term	Magnitude	Type	Enters τ_{eff}?
Calibration	±5%	systematic	cancels in ratio
Non-linearity	±1%	systematic	cancels (1st order)
Repeatability	±0.5%	random	yes
Cosine/tilt	±2% (±5% SZA$>72^\circ$); tilt p95 2.6°	random (swing)	yes
Temperature response	$\leq$0.1% °C^{-1} × $\lvert T-25\rvert$	systematic	partially
Quantisation (0.01 V)	0.15–7%	random	yes
Altitude (BME280, ±1 hPa)	±8.7 m (surface) to ±96 m (apogee)	systematic	negligible (500-m bins)
GM counting	Poisson $\sqrt{N}$	random	—

3.2 Effective optical depth

Samples are averaged in 500-m altitude bins (ascent only). Modelling global irradiance with a Beer–Lambert form,

$$\ln E = \ln E_0 + \ln\cos\theta_s - m(\theta_s)\,\tau_{\text{eff}}(z), \tag{2}$$

where $m(\theta_s)$ is the Kasten–Young relative air mass [13] and $\tau_{\text{eff}}(z)$ is the *effective* vertical optical depth of the column above altitude z, we anchor the retrieval at apogee with $\tau_{\text{eff}}(z_{\text{apo}}) = \tau_R(360)\,P_{\text{apo}}/P_0 + 0.005 = 0.050$ (residual Rayleigh above 81 hPa plus stratospheric aerosol). The Rayleigh optical depth uses $\tau_R(\lambda) = 0.0088\,\lambda^{-4.05}\,P/P_0$ (λ in μm) [14], giving $\tau_R = 0.551$ at the sensor's effective wavelength (≈360 nm). Because the anchor and every sample share the same calibration factor, the ±5% calibration and non-linearity terms cancel in Eq. (2); the propagated uncertainty of τ_{eff} retains only the random terms of the two bins involved:

$$\sigma_\tau(z) = \frac{1}{m(\theta_s)}\sqrt{\sigma_{\ln E}^2(z) + \sigma_{\ln E}^2(z_{\text{apo}})}. \tag{3}$$

The implied extraterrestrial constant, $E_0 = 102\ \mathrm{W\,m^{-2}}$, is $\sim$20% above the nominal top-of-atmosphere UV-A irradiance for this band ($\approx$85 $\mathrm{W\,m^{-2}}$), consistent with the combined calibration, spectral-band and tilt systematics discussed in Section 6; this offset does not affect τ_{eff}.

3.3 Satellite comparison quantities

The Ångström exponent [15] from the along-track MAIAC pair is $\alpha = \ln(0.781/0.610)/\ln(550/470) = 1.57$, giving $\mathrm{AOD}_{360} = 0.610\,(550/360)^{1.57} = 1.19 \pm 0.21$, where the uncertainty combines the MAIAC expected error over land ($\pm 0.05 \pm 0.1\,\mathrm{AOD}$) and a 10% extrapolation allowance. The direct-beam extinction expectation is then $\tau_{\mathrm{ext}} = \tau_R + \mathrm{AOD}_{360} = 1.74 \pm 0.21$.

4 Results

4.1 Profiles

Figure 2 shows the flight profile and payload-internal thermodynamic record. Figure 3a shows measured UV-A irradiance increasing from 7.6 to 79.2 $\mathrm{W\,m^{-2}}$ between the surface and apogee. Most of that increase is solar geometry: after normalising by $\cos\theta_s$ (Fig. 3b) the profile is nearly constant above $\approx$3 km and drops only in the lowest 2–3 km, where the aerosol column resides.

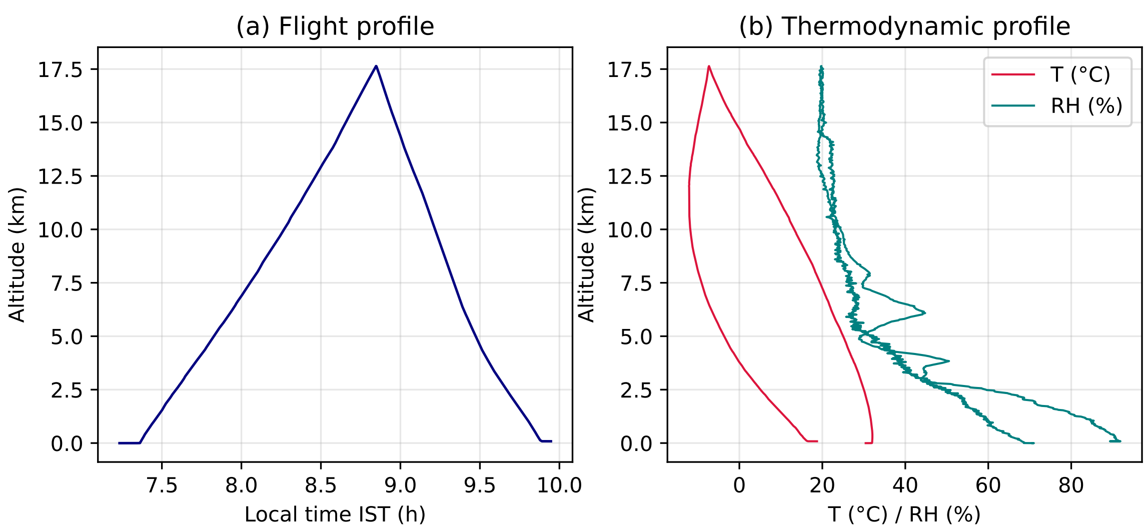

Figure 2: (a) Altitude–time record (RTC-anchored IST). (b) Payload-internal temperature and relative-humidity profiles.

4.2 Effective optical depth of global UV-A

The Langley-anchored retrieval (Fig. 4) yields a surface value

$$\tau_{\mathrm{eff}} = 0.233 \pm 0.023,$$

i.e. a vertical-equivalent transmittance of global UV-A of $e^{-\tau_{\mathrm{eff}}} = 0.79$. Roughly 90% of the column effect accumulates below $\approx$2.5 km, with the sharpest decrease in the 1.5–2.5 km layer — at and just above the ERA5 boundary-layer top (1.41 km) and cloud base (1.29 km) — consistent with aerosol and scattered cloud accumulating at the top of the shallow morning boundary layer. The lowest bins were measured at SZA $> 72^\circ$ and constrain the sub-1.5-km contribution only weakly.

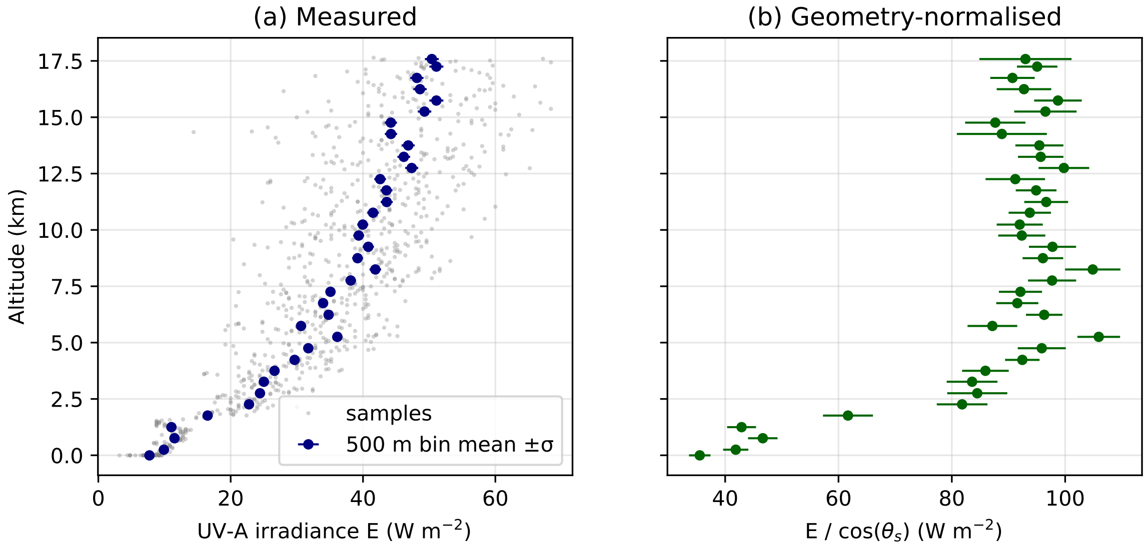

Figure 3: (a) Global UV-A irradiance vs. altitude: individual 7-s samples (grey) and 500-m bin means with $\pm 1\sigma$ random uncertainty (blue). (b) The same bins normalised by $\cos\theta_s$; the residual altitude dependence below $\approx$3 km is atmospheric attenuation.

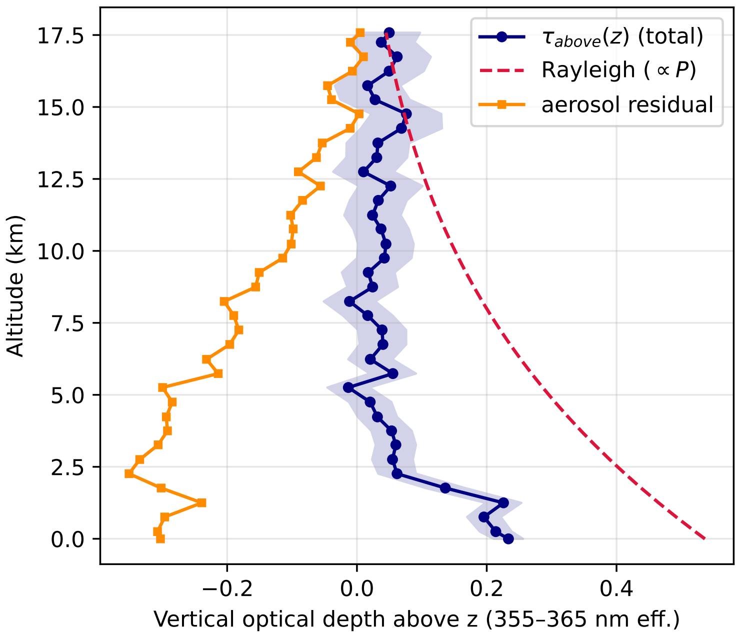

Figure 4: Effective vertical optical depth above altitude z for global UV-A (blue, with $\pm 1\sigma$ envelope), the Rayleigh extinction profile (dashed), and their difference (orange). τ_{eff} falling *below* the Rayleigh extinction curve is the signature of diffuse replenishment: scattering removes photons from the direct beam but returns most of them to the downwelling hemisphere, which is what a 180° field-of-view radiometer measures.

4.3 Global vs. direct-beam attenuation

Figure 5 compares the measured τ_{eff} with the direct-beam expectation. The ratio

$$\eta = \frac{\tau_{\text{eff}}}{\tau_{\text{ext}}} = 0.134 \pm 0.021$$

states that the real atmosphere on this day attenuated hemispherical UV-A only ≈13% as strongly as it attenuated the direct beam. Two independent observations support the interpretation that scattering (not absorption) dominated the extinction: the TROPOMI UV aerosol index was slightly negative (−0.19, non-absorbing), and surface $PM_{2.5}$ was moderate (24–29 $\mu g\,m^{-3}$). For a purely scattering, high-single-scattering-albedo column, most extinguished direct-beam photons reappear as diffuse sky radiance, so $\eta \ll 1$ is the physically expected regime; η would approach unity only for a strongly absorbing column.

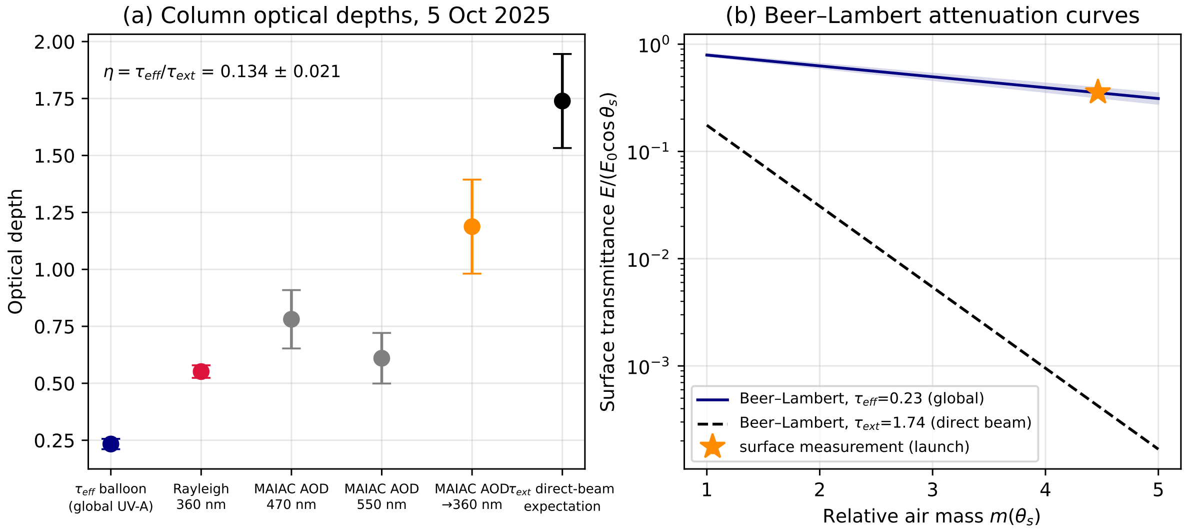

Figure 5: (a) Optical-depth comparison for 5 October 2025 over the Kota region. Blue: balloon-derived effective optical depth for global UV-A. Red: Rayleigh extinction at 360 nm. Grey: MAIAC along-track AOD at 470 and 550 nm. Orange: MAIAC AOD extrapolated to 360 nm with $\alpha = 1.57$. Black: direct-beam extinction expectation (Rayleigh + aerosol). (b) The same result as Beer–Lambert attenuation curves: surface transmittance vs. air mass for the measured effective optical depth (solid, with $\pm 1\sigma$ band) and for the direct-beam expectation (dashed). The pre-launch surface measurement (star) lies on the global-irradiance curve; the direct-beam curve at that air mass predicts a transmittance ≈900× smaller, which is the clearest visual statement that a hemispherical sensor under a scattering atmosphere is dominated by diffuse irradiance.

4.4 Spatial representativeness of the satellite AOD

MAIAC 550-nm AOD statistics in growing buffers around the launch site show a systematic dilution with distance: mean 0.664 (±0.038) within 10 km, 0.635 (±0.103) within 25 km, 0.554 (±0.125) within 50 km, and 0.461 (±0.102) within 100 km. The along-track value (0.610) sits between the 10 and 25 km buffer means, indicating the flight sampled air representative of the urban core rather than the cleaner surroundings — a scale effect consistent with validation practice for satellite AOD [6, 3].

4.5 Context: surface air quality and cosmic-ray counts

Figure 6a places the flight within the CPCB record: daily-mean $PM_{2.5}$ on launch day was 24.2 $\mu g\,m^{-3}$ (Nayapura) and 28.6 $\mu g\,m^{-3}$ (Shrinath Puram), with 31.6 and 28.2 $\mu g\,m^{-3}$ in the 08:00–12:00 window — a moderate pollution day, well below the post-Diwali November peaks visible later in the same record. The GM channel (Fig. 6b) shows the expected secondary cosmic-ray increase from 30 CPM at the surface to a median 72 CPM at 4–4.5 km (×2.4, Poisson $\sigma = \sqrt{N}$), before failing above 4.7 km (Section 5).

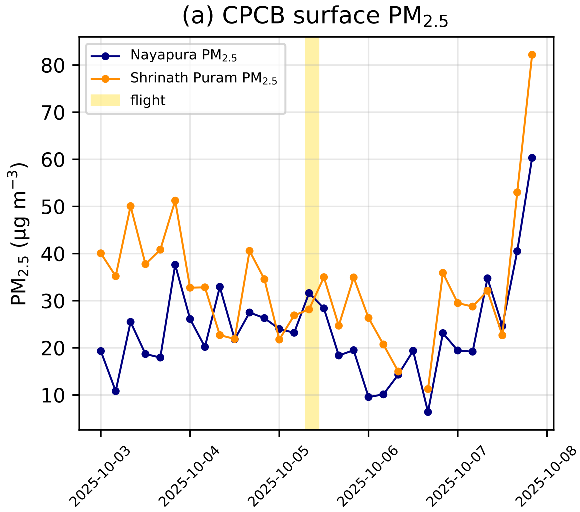

Figure 6: (a) CPCB 4-hourly $PM_{2.5}$ at the two Kota stations, 3–8 October 2025; the gold band marks the flight. (b) GM count-rate profile over the valid segment (15-point median $\pm\sqrt{N}$); data above the high-voltage failure at ≈4.7 km are excluded.

4.6 Independent surface cross-check against NASA POWER

The radiometer can be referenced against an independent, satellite-derived estimate of surface UV-A. The NASA POWER hourly product (SYN1DEG; 1° resolution, 315–400 nm band) for the launch coordinates gives, interpolated to the pre-launch epoch (07:19 IST), 6.5 $\mathrm{W\,m^{-2}}$; the level, ground-based SU-202 read 8.0 $\mathrm{W\,m^{-2}}$ — agreement within ≈25%, which is consistent with the combined effect of the 1° footprint, the steep sunrise ramp within the product's hourly averaging window, the slight band mismatch (315–400 vs. 305–390 nm), and the sensor's low-signal quantisation at that hour. After touchdown (09:54 IST) the sensor read 11.7 $\mathrm{W\,m^{-2}}$ against a POWER value of 35.6 $\mathrm{W\,m^{-2}}$: the landed payload no longer viewed the full sky, and all post-touchdown data are accordingly excluded from analysis — a reminder that orientation control ends at landing, not at descent.

5 Instrument lessons

Three failures on this flight are instructive for HAB payload design and are reported so that others can avoid them:

1. **GM-tube corona discharge at low pressure.** Above ≈4.7 km (P $\lesssim$ 550 hPa) the count rate jumped from $\sim 10^2$ to $>10^5$ CPM and then fell to zero: the high-voltage supply arced in thin air. Counts above 4.7 km were discarded. Conformal coating or a sealed tube enclosure is recommended.

2. **Non-responsive O_3 channel.** The semiconductor ozone sensor read 0.00 throughout; heater-type metal-oxide gas sensors need minutes of warm-up power and ambient temperatures they never see on ascent. Electrochemical cells or pre-heated enclosures are preferable.

3. **GPS time-stamp freezing.** Position remained usable but time strings froze intermittently at altitude. Anchoring the RTC to the first valid GPS fix recovered a reliable timeline (verified by a physical 3.0 $\mathrm{m\,s^{-1}}$ mean ascent rate and by the IMU event chain); logging both clocks is essential because solar-geometry corrections are only as good as the timeline.

One positive lesson balances the three failures: an ordinary action camera with gyro logging (intended for video stabilisation) doubles as a free 1-kHz attitude sensor. It quantified payload tilt and rotation, independently confirmed the event timeline, and documented sky conditions — capabilities that would otherwise require a dedicated IMU payload. We recommend gyro-logging cameras as standard equipment on radiometric HAB flights.

6 Limitations

This is a single flight with a hemispherical broadband sensor, and four caveats bound the interpretation. (i) τ_{eff} is an *effective* (global-irradiance) quantity, not extinction AOD; the Beer–Lambert form of Eq. (2) is an operational definition, and the plane-parallel air-mass treatment of the diffuse component is approximate, particularly at the large launch SZA. Moreover, the morning was partly cloudy below $\approx$3 km, so τ_{eff} is an all-sky effective quantity while the MAIAC AOD it is compared against is cloud-screened; part of the boundary-layer-concentrated attenuation may be broken-cloud rather than aerosol extinction. (ii) Payload tilt under balloon swing is now measured (median 0.9°, 95th percentile 2.6°, with $\approx$6.8 rpm rotation averaging the first-order cosine error), so this term is bounded rather than assumed. (iii) The effective wavelength of the 305–390 nm band ($\approx$360 nm) carries uncertainty that maps onto τ_R and the Ångström extrapolation. (iv) The implied E_0 exceeds the nominal top-of-atmosphere value by $\sim$20%, indicating residual absolute-scale systematics (calibration transfer, spectral response, tilt); ratio-based quantities (τ_{eff}, η) are insensitive to this scale. A repeat flight near local noon, a shaded/global sensor pair, or an added narrowband photodiode would separate the direct and diffuse components directly.

7 Conclusions

A low-cost HAB flight with one research-grade radiometer and rigorous error propagation produced a quantitative, uncertainty-bounded result: on a moderately turbid, non-absorbing day over Kota, the vertical effective optical depth of global UV-A was 0.233 ± 0.023 — only 13.4 ± 2.1% of the direct-beam extinction expectation constructed from Rayleigh theory and satellite AOD. The attenuation was concentrated below $\approx$2.5 km, chiefly in the layer at and just above the reanalysis boundary-layer top and cloud base. The flight also demonstrated, through three documented failures, where student-class payloads are fragile. The combination of a calibrated primary sensor, cancellation-aware uncertainty analysis, and free public satellite/reanalysis/regulatory data is reproducible by any school or university group, and repeat flights across seasons would turn this case study into a useful climatology of UV attenuation for the region.

Data and code availability

The flight datalog, cleaned profiles, analysis scripts, and figure-generation code are available at `https://github.com/1boscochiramel/AkashaJyoti-UVA-Kota`.

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Conflicts of interest

The author declares no conflicts of interest.

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